Hints: Abstract algebra – 4. Group actions, homomorphisms and automorphisms

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Two easy facts: $N_{UVW} = N_{VWU} = N_{WUV}$, since $xyz = e \iff yzx = e$; and $N_{UVG} = |U|\,|V|$, since z is determined by x, y.

Hint 2. Expand $|A|\,|B| = N_{ABG} = N_{ABA} + N_{ABB} + N_{ABC}$ and $|B|\,|A| = N_{BAG} = N_{BAA} + N_{BAB} + N_{BAC}$. The first two terms match by cyclic invariance, so $N_{ABC} = N_{BAC}$, which is N_{CBA} by cyclic invariance.

Problem 2

Hint 1. Let $T = \{gh, gh^{-1}, g^{-1}h, g^{-1}h^{-1}\}$. You must show that every element is a product of between 1 and $|G|$ elements of T. Let S be the set of all finite products of elements of T. In a finite group, a set closed under multiplication is a subgroup.

Hint 2. S contains $(gh)(g^{-1}h)^{-1} = g^2$. Since $\operatorname{ord}(g)$ is odd, $g \in \langle g^2\rangle \subseteq S$, then $h = g^{-1}(gh) \in S$. So $S = G$.

Hint 3. For the bound, let U_r be the set of products of at most r elements of T, so $U_{r+1} = T \cup U_r T$. If $U_{r+1} = U_r$, the chain is stationary from then on. Before that it grows strictly, starting from $|U_1| \geqslant 1$, so $U_{|G|} = S = G$.

Problem 3

Hint 1. An endomorphism is determined by the images of a generating set. Choose generators greedily, $g_{i+1} \notin \langle g_1, \ldots, g_i\rangle$. Each step raises the order of the generated subgroup by a factor that is at least a prime, so G has a generating set with $k \leqslant \Omega(n)$ elements, where Ω counts prime factors with multiplicity.

Hint 2. So $|\operatorname{End} G| \leqslant n^k \leqslant n^{\Omega(n)}$. Write $n = pm$; then $\Omega(n) = 1 + \Omega(m) \leqslant 1 + (m-1) = n/p$, because $\Omega(m) \leqslant \log_2 m \leqslant m - 1$.

Hint 3. Equality needs $\Omega(m) = m - 1$, i.e. $m \in \{1, 2\}$, and every choice of images of the generators must extend. So $n = p$ or $n = 2p$ (with p the largest prime). Check the candidates: $\mathbb{Z}/p$ has $p = n^1$ endomorphisms; for $n = 4$, $(\mathbb{Z}/2)^2$ has $16 = 4^2$ and $\mathbb{Z}/4$ only 4; for $n = 2p$ with p odd, $\mathbb{Z}/2p$ has $2p$ and D_p has $p^2 + 1$, both less than $4p^2$. Equality: $\mathbb{Z}/p$ and $(\mathbb{Z}/2)^2$.

Problem 4

Hint 1. Let $d = \gcd(m, n) = sm + tn$, $A = G(m)$, $B = G(n)$. Both are normal (generated by conjugation-invariant sets). Since $x^d = (x^m)^s (x^n)^t$, you get $G(d) = \langle A, B\rangle$.

Hint 2. For $a \in A$, $b \in B$ the commutator $[a, b]$ lies in $A \cap B$, by normality of each. Elements of $A \cap B$ commute with A and with B, hence are central in $H = G(d)$. So H has nilpotency class at most 2, and the commutator is bimultiplicative on H.

Hint 3. Let $p = x^d$, $q = y^d$, and $m = dm'$, $n = dn'$. Then $p^{m'}, q^{m'} \in A$ commute, so $[p, q]^{m'^2} = e$; likewise $[p, q]^{n'^2} = e$. Since $\gcd(m'^2, n'^2) = 1$, $[p, q] = e$. So the generators of H commute.

Problem 5

Hint 1. An endomorphism of $\mathbb{Z}/n$ is $x \mapsto ax$ with $a = f(1)$, and it is bijective iff $\gcd(a, n) = 1$. An additive map of $(\mathbb{Z}/p)^k$ is $\mathbb{F}_p$-linear. $|\mathrm{GL}_k(\mathbb{F}_p)| = \prod_{i=0}^{k-1}(p^k - p^i)$: choose the columns one by one outside the span of the previous ones.

Hint 2. If $|\operatorname{Aut} G| = 1$, then $\operatorname{Inn} G \cong G/Z(G)$ is trivial, so G is abelian. Then $x \mapsto x^{-1}$ is an automorphism, so $x^2 = e$ for all x.

Hint 3. So $G \cong (\mathbb{Z}/2)^k$ with $\mathrm{GL}_k(\mathbb{F}_2)$ trivial, i.e. $k \leqslant 1$: G is trivial or $\mathbb{Z}/2$.

Problem 6

Hint 1. The rotation group of the cube has 24 elements: the identity; 6 rotations by $\pm 90^\circ$ about face axes; 3 by 180° about face axes; 8 by $\pm 120^\circ$ about vertex axes; 6 by 180° about edge axes. For each type, count the cycles on faces and on vertices.

Hint 2. Faces: $\frac{1}{24}(k^6 + 3k^4 + 12k^3 + 8k^2)$. Vertices: $\frac{1}{24}(k^8 + 17k^4 + 6k^2)$. Check: for $k = 2$ these give 10 and 23.

Hint 3. Integrality follows from Burnside: the expression counts orbits. A polynomial with non-integral coefficients can still be integer-valued, like $\binom{k}{2}$.

Problem 7

Hint 1. Let e be the identity of $(G, *)$. For any a, $a * e = a$, while $a \times e \perp a$. So either $a = 0$ or $a \times e = 0$. If $e \neq 0$, every element is parallel to e and all cross products vanish. So assume $e = 0$.

Hint 2. Suppose $a \times b \neq 0$, so $c = a * b = a \times b$. Compute $(a * a) * b = a * c = a \times c$ and $(a * a) * c = a * (a \times c) = a \times (a \times c) = -|a|^2 c$. Neither is orthogonal to b, resp. c.

Hint 3. So $(a * a) \times b = 0$ and $(a * a) \times c = 0$, i.e. $a * a$ is parallel to both b and c, hence $a * a = 0 = e$. Then $b = (a * a) * b = a \times c = a \times (a \times b)$, which forces $a \perp b$ and $b = -|a|^2 b$: impossible.

Problem 8

Hint 1. Let $F(g)$ be the number of fixed points and $n = |X|$. Burnside gives $\sum_g F(g) = |G|$. Applying it to the action on $X \times X$ (at least two orbits: the diagonal and its complement) gives $\sum_g F(g)^2 \geqslant 2|G|$.

Hint 2. Consider $\sum_g (F(g) - 1)(F(g) - n)$. It equals $\sum F^2 - |G| \geqslant |G|$. Each term with $1 \leqslant F(g) \leqslant n$ is $\leqslant 0$, and each g without fixed points contributes exactly n.

Hint 3. Hence $n \cdot \#\{g : F(g) = 0\} \geqslant |G|$, so the proportion is at least $1/n$, in particular positive.

Problem 9

Hint 1. $\varphi \circ i_g \circ \varphi^{-1} = i_{\varphi(g)}$.

Hint 2. Two transpositions commute iff they are disjoint (otherwise their product has order 3). So $\varphi(1\,2), \ldots, \varphi(1\,n)$ are pairwise non-commuting transpositions. For $n \geqslant 4$, show they all share one common point a (three pairwise intersecting transpositions either share a point or form a triangle $(xy), (yz), (xz)$, and a fourth one cannot meet all three sides of a triangle).

Hint 3. Write $\varphi(1\,i) = (a\,b_i)$ and define $\sigma(1) = a$, $\sigma(i) = b_i$. Then φ and conjugation by σ agree on the generators $(1\,i)$. ($n \leqslant 3$: check directly.)

Problem 10

Hint 1. The action on the n cosets gives $G \to S_n$. Its kernel K is normal, and G/K embeds in S_n, so $[G : K]$ divides $n!$.

Hint 2. If a simple group of order 60 had a subgroup of index k with $1 < k < 5$, the kernel would be a proper normal subgroup, hence trivial. Then G would embed in S_k, and $60 \mid k!$ is impossible for $k \leqslant 4$.